Practice Set: Higher Engineering Mathematics

Section 1: Differential Calculus – Maxima, Minima & Lagrange Multipliers

1. For a function $f(x, y)$ to have a stationary point at (a, b), which of the following conditions must hold?
(A) $f_x(a, b) = 0$ and $f_y(a, b) = 0$ (B) $f_x(a, b) > 0$ and $f_y(a, b) < 0$
(C) $f_{xx}(a, b) = 0$ only (D) $f_{xy}(a, b) = 0$ only

2. Let $r = f_{xx}$, $s = f_{xy}$, and $t = f_{yy}$ evaluated at a critical point (a, b). The function $f(x, y)$ achieves a local minimum at (a, b) if:
(A) $rt - s^2 > 0$ and $r > 0$ (B) $rt - s^2 > 0$ and $r < 0$
(C) $rt - s^2 < 0$ and $r > 0$ (D) $rt - s^2 = 0$ and $r > 0$

3. Under the second-derivative test for $f(x, y)$, a local maximum occurs at (a, b) if:
(A) $rt - s^2 > 0$ and $r < 0$ (B) $rt - s^2 > 0$ and $r > 0$
(C) $rt - s^2 < 0$ and $r < 0$ (D) $rt - s^2 = 0$ and $r < 0$

4. If $rt - s^2 < 0$ at a critical point (a, b) of $f(x, y)$, then (a, b) is classified as a:
(A) Point of local minimum (B) Point of local maximum
(C) Saddle point (D) Boundary point

5. When $rt - s^2 = 0$ at a critical point (a, b), the second derivative test is:
(A) Conclusive for maximum (B) Conclusive for minimum
(C) Inconclusive (D) Guaranteeing a saddle point

6. In Lagrange's method of undetermined multipliers for optimizing $f(x, y)$ subject to constraint $\phi(x, y) = 0$, the auxiliary function $F(x, y, \lambda)$ is defined as:
(A) $F = f + \lambda\phi$ (B) $F = f \cdot \phi + \lambda$
(C) $F = \frac{f}{\phi} + \lambda$ (D) $F = f - \lambda^2\phi$

7. To find the stationary points of the auxiliary function $F(x, y, \lambda) = f(x, y) + \lambda\phi(x, y)$, one must solve the system:
(A) $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial \lambda} = 0$ (B) $\frac{\partial F}{\partial x} = 1, \frac{\partial F}{\partial y} = 1, \frac{\partial F}{\partial \lambda} = 0$
(C) $f_x + f_y = 0, \phi_x + \phi_y = 0$ (D) $\frac{\partial^2 F}{\partial x^2} = 0, \frac{\partial^2 F}{\partial y^2} = 0$

8. If $f(x, y, z)$ is subject to two constraints $\phi_1(x, y, z) = 0$ and $\phi_2(x, y, z) = 0$, Lagrange's auxiliary function F is constructed as:
(A) $F = f + \lambda_1\phi_1 + \lambda_2\phi_2$ (B) $F = f + \lambda_1\phi_1 \cdot \lambda_2\phi_2$
(C) $F = f \cdot (\lambda_1\phi_1 + \lambda_2\phi_2)$ (D) $F = \lambda_1 f + \lambda_2\phi_1 + \phi_2$

9. The absolute maximum and absolute minimum of a continuous function on a closed bounded region D can occur at:
(A) Interior critical points or boundary points
(B) Strictly interior points only
(C) Strictly boundary points only
(D) Points where $rt - s^2 < 0$ only

10. In Lagrange's auxiliary function $F = f + \lambda\phi$, the parameter λ represents:
(A) The rate of change of the optimal value with respect to the constraint bound
(B) The curvature of f
(C) The maximum value of f
(D) The distance from origin

11. For $f(x, y) = x^2 + y^2$, the origin $(0, 0)$ is a:
(A) Local minimum (B) Local maximum (C) Saddle point (D) Test inconclusive

12. For $f(x, y) = x^2 - y^2$, the origin $(0, 0)$ is a:
(A) Local minimum (B) Local maximum (C) Saddle point (D) Absolute maximum

13. For $f(x, y) = -x^2 - y^2$, the origin $(0, 0)$ is a:
(A) Local minimum (B) Local maximum (C) Saddle point (D) Inconclusive point

14. If $F(x, y, \lambda) = f(x, y) + \lambda\phi(x, y)$, setting $\frac{\partial F}{\partial \lambda} = 0$ recovers:
(A) The constraint equation $\phi(x, y) = 0$ (B) The objective function $f(x, y) = 0$
(C) The multiplier $\lambda = 0$ (D) The derivative $f_x = 0$

15. Find the critical point of $f(x, y) = 2x^2 + 5y^2 - 8x - 20y + 14$.
(A) $(2, 2)$ (B) $(1, 2)$ (C) $(4, 2)$ (D) $(0, 0)$

16. Evaluate the minimum value of $f(x, y) = 2x^2 + 5y^2 - 8x - 20y + 14$.
(A) -14 (B) 14 (C) 0 (D) -28

17. Find the critical points of $f(x, y) = x^3 + y^3 - 12xy$.
(A) $(0, 0)$ and $(4, 4)$ (B) $(0, 0)$ and $(2, 2)$ (C) $(1, 1)$ and $(-1, -1)$ (D) $(3, 3)$ only

18. For $f(x, y) = x^3 + y^3 - 12xy$, classify the critical point $(4, 4)$.
(A) Local minimum (B) Local maximum (C) Saddle point (D) Test inconclusive

19. For $f(x, y) = x^3 + y^3 - 12xy$, classify the critical point $(0, 0)$.
(A) Saddle point (B) Local minimum (C) Local maximum (D) Absolute minimum

20. Find the stationary points of $f(x, y) = 3x^2 - y^2 + 12x - 6y + 2$.
(A) $(-2, -3)$ (B) $(2, 3)$ (C) $(-2, 3)$ (D) $(0, 0)$

21. Classify the critical point $(-2, -3)$ of $f(x, y) = 3x^2 - y^2 + 12x - 6y + 2$.
(A) Saddle point (B) Local minimum (C) Local maximum (D) Inconclusive

22. Find the local extreme value of $f(x, y) = x^2 + 4y^2 - 6x - 16y + 20$.
(A) -5 at $(3, 2)$ (B) 5 at $(3, 2)$ (C) 0 at $(0, 0)$ (D) -10 at $(2, 3)$

23. Calculate r, s, t for $f(x, y) = x^4 + y^4 - 4xy$ at $(1, 1)$.
(A) $r = 12, s = -4, t = 12$ (B) $r = 4, s = -4, t = 4$ (C) $r = 12, s = 0, t = 12$ (D) $r = 3, s = -1, t = 3$

24. Evaluate $rt - s^2$ at $(1, 1)$ for $f(x, y) = x^4 + y^4 - 4xy$.
(A) $128 > 0$ (B) $-16 < 0$ (C) 0 (D) 144

25. Determine the nature of $(1, 1)$ for $f(x, y) = x^4 + y^4 - 4xy$.
(A) Local minimum (B) Local maximum (C) Saddle point (D) Test fails

26. For $f(x, y) = x^4 + y^4 - 4xy$, determine the nature of $(0, 0)$.
(A) Saddle point (B) Local minimum (C) Local maximum (D) Absolute minimum

27. Find the minimum value of $f(x, y) = x^2 + y^2 + \frac{16}{x} + \frac{16}{y}$ for $x, y > 0$.
(A) 24 at $(2, 2)$ (B) 16 at $(2, 2)$ (C) 32 at $(4, 4)$ (D) 12 at $(1, 1)$

28. Find the critical point of $f(x,y) = x^2 + 3xy + y^2 - 5x - 5y + 3$.
(A) $(1,1)$ (B) $(2,2)$ (C) $(-1,-1)$ (D) $(0,5)$

29. Classify the critical point $(1,1)$ of $f(x,y) = x^2 + 3xy + y^2 - 5x - 5y + 3$.
(A) Saddle point since $rt - s^2 = 4 - 9 = -5 < 0$ (B) Local minimum
(C) Local maximum (D) Inconclusive

30. Using $F = xy + \lambda(2x + 3y - 12)$, find the maximum of $f(x,y) = xy$ subject to $2x + 3y = 12$.
(A) 6 at $(3,2)$ (B) 12 at $(3,2)$ (C) 8 at $(2,3)$ (D) 4 at $(3,2)$

31. In $F = xy + \lambda(2x + 3y - 12)$, what is the value of λ at the stationary point?
(A) -1 (B) $-\frac{3}{2}$ (C) -2 (D) $-\frac{2}{3}$

32. Minimize $f(x,y) = x^2 + y^2$ subject to $3x + 4y = 25$ using $F = x^2 + y^2 + \lambda(3x + 4y - 25)$.
(A) Minimum value 25 at $(3,4)$ (B) Minimum value 50 at $(6,8)$
(C) Minimum value 16 at $(4,3)$ (D) Minimum value 5 at $(3,4)$

33. In minimizing $f(x,y) = x^2 + y^2$ subject to $3x + 4y = 25$, the value of λ is:
(A) -2 (B) 2 (C) -5 (D) -1

34. Maximize $f(x,y,z) = xyz$ subject to $x + y + z = 18$ $(x,y,z > 0)$ using $F = xyz + \lambda(x + y + z - 18)$.
(A) 216 at $(6,6,6)$ (B) 125 at $(5,5,8)$ (C) 512 at $(8,8,2)$ (D) 64 at $(4,4,10)$

35. Find the stationary point of $F = (x^2 + y^2 + z^2) + \lambda(x + 2y + 3z - 14)$.
(A) $(1,2,3)$ (B) $(2,2,2)$ (C) $(3,2,1)$ (D) $(0,1,4)$

36. Maximize $f(x,y) = x^3y^2$ subject to $x + y = 10$ $(x,y > 0)$ using $F = x^3y^2 + \lambda(x + y - 10)$.
(A) 3456 at $(6,4)$ (B) 2000 at $(5,5)$ (C) 1728 at $(4,6)$ (D) 512 at $(8,2)$

37. Optimize $f(x,y) = 5x + 12y$ subject to $x^2 + y^2 = 169$ using $F = 5x + 12y + \lambda(x^2 + y^2 - 169)$. The maximum value is:
(A) 169 at $(5,12)$ (B) 130 at $(5,12)$ (C) 225 at $(9,12)$ (D) 65 at $(5,0)$

38. In optimizing $f(x,y) = 5x + 12y$ subject to $x^2 + y^2 = 169$, the multiplier λ for maximum is:
(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) -1 (D) $-\frac{1}{13}$

39. Dimensions of an open rectangular box of volume $V = 108$ having minimum surface area $S = xy + 2xz + 2yz$ using $F = xy + 2xz + 2yz + \lambda(xyz - 108)$ are:
(A) $x = 6, y = 6, z = 3$ (B) $x = 3, y = 3, z = 12$ (C) $x = 4, y = 4, z = 6.75$ (D) $x = 9, y = 4, z = 3$

40. Minimum surface area of the open box with $V = 108$ is:
(A) 108 (B) 72 (C) 144 (D) 216

41. Maximum value of $f(x,y,z) = x + 2y + 3z$ subject to $x^2 + y^2 + z^2 = 14$ using $F = (x + 2y + 3z) + \lambda(x^2 + y^2 + z^2 - 14)$ is:
(A) 14 (B) 28 (C) $\sqrt{14}$ (D) 42

42. Find the point on $x^2 + y^2 + z^2 = 14$ furthest from origin along vector $(1,2,3)$.
(A) $(1,2,3)$ (B) $(-1,-2,-3)$ (C) $(2,1,3)$ (D) $(3,2,1)$

43. Determine the critical points of $f(x,y) = x^3 - 3x + y^2 - 4y + 5$.
(A) $(1,2)$ and $(-1,2)$ (B) $(0,0)$ and $(1,1)$ (C) $(2,1)$ and $(-2,1)$ (D) $(3,4)$

44. Classify $(1,2)$ for $f(x,y) = x^3 - 3x + y^2 - 4y + 5$.
(A) Local minimum (B) Local maximum (C) Saddle point (D) Inconclusive

45. Classify $(-1,2)$ for $f(x,y) = x^3 - 3x + y^2 - 4y + 5$.
(A) Saddle point (B) Local minimum (C) Local maximum (D) Test fails

46. For $F = (x^2 + 3y^2 + 2z^2) + \lambda(x + y + z - 11)$, the ratios $x : y : z$ are:
(A) $1 : \frac{1}{3} : \frac{1}{2}$ (B) $1 : 3 : 2$ (C) $3 : 1 : 2$ (D) $1 : 1 : 1$

47. Value of x for stationary point of $F = (x^2 + 3y^2 + 2z^2) + \lambda(x + y + z - 11)$ is:
(A) 6 (B) 3 (C) 2 (D) 1

48. For $f(x, y) = e^{-(x^2+y^2)}$, the origin $(0, 0)$ is a:
(A) Absolute maximum with value 1 (B) Absolute minimum with value 0
(C) Saddle point (D) Inconclusive

49. Function $f(x, y) = 2x^2 - 4xy + y^2$ at origin $(0, 0)$ has:
(A) Saddle point ($rt - s^2 = 4 - 16 = -12 < 0$) (B) Local minimum
(C) Local maximum (D) Inconclusive

50. Maximum of $f(x, y) = 6x + 8y$ subject to $x^2 + y^2 = 100$ is:
(A) 100 (B) 140 (C) 50 (D) 200

51. Minimum distance from origin to $2x + y + 2z = 18$ using $F = (x^2 + y^2 + z^2) + \lambda(2x + y + 2z - 18)$ is:
(A) 6 (B) 18 (C) 36 (D) 3

Section 2: Linear Algebra – Matrix Rank, Inverses, Normal Form & Linear Systems

1. The rank of a matrix A is r if:
(A) At least one minor of order r is non-zero, and all minors of order $r + 1$ are zero
(B) All minors of order r are non-zero
(C) The matrix is square of order r
(D) The determinant of order r is zero

2. If A is an $m \times n$ matrix, then $\text{rank}(A)$ satisfies:
(A) $\text{rank}(A) \leq \min(m, n)$ (B) $\text{rank}(A) \geq \max(m, n)$
(C) $\text{rank}(A) = m + n$ (D) $\text{rank}(A) = m \cdot n$

3. Which elementary transformation alters the rank of a matrix?
(A) Interchanging two rows (B) Multiplying a row by a non-zero scalar
(C) Adding a non-zero scalar multiple of one row to another (D) None of these

4. If A^T is the transpose of matrix A, then:
(A) $\text{rank}(A^T) = \text{rank}(A)$ (B) $\text{rank}(A^T) \neq \text{rank}(A)$
(C) $\text{rank}(A^T) = \frac{1}{\text{rank}(A)}$ (D) $\text{rank}(A^T) = 0$

5. For matrices A and B compatible for addition, subadditivity of rank states:
(A) $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$ (B) $\text{rank}(A + B) = \text{rank}(A) + \text{rank}(B)$
(C) $\text{rank}(A + B) \geq \text{rank}(A) + \text{rank}(B)$ (D) $\text{rank}(A + B) = \text{rank}(A) \cdot \text{rank}(B)$

6. Sylvester's Law of Nullity for matrices $A_{m\times n}$ and $B_{n\times p}$ states:
(A) $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$ (B) $\text{rank}(AB) \leq \text{rank}(A) + \text{rank}(B) - n$
(C) $\text{rank}(AB) = \text{rank}(A) + \text{rank}(B)$ (D) $\text{rank}(AB) = \text{rank}(A) - \text{rank}(B)$

7. An $n \times n$ square matrix A is non-singular if and only if:
(A) $\text{rank}(A) = n$ (B) $\text{rank}(A) < n$
(C) $\text{rank}(A) = 0$ (D) $\det(A) = 0$

8. Every non-singular matrix A of order n can be reduced by elementary operations to:
(A) I_n (B) 0_n (C) Diagonal matrix with zeros on main diagonal (D) Upper triangular with zero diagonal

9. The normal form of an $m \times n$ matrix of rank r is:
(A) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & I_r \\ I_r & 0 \end{bmatrix}$
(C) $\begin{bmatrix} I_r & I_r \\ 0 & 0 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

10. The homogeneous system $AX = 0$ with n unknowns always possesses:
(A) At least the trivial solution $X = 0$ (B) No solution
(C) Strictly non-trivial solutions only (D) An inconsistent state

11. The system $AX = 0$ has a non-trivial solution if and only if:
(A) $\text{rank}(A) < n$ (B) $\text{rank}(A) = n$
(C) $\det(A) \neq 0$ (D) $\text{rank}(A) > n$

12. The system $AX = B$ is consistent if and only if:
(A) $\text{rank}(A) = \text{rank}([A|B])$ (B) $\text{rank}(A) \neq \text{rank}([A|B])$
(C) $\text{rank}(A) > \text{rank}([A|B])$ (D) $\text{rank}([A|B]) = n + 1$

13. The non-homogeneous system $AX = B$ with n unknowns has a unique solution if and only if:
(A) $\text{rank}(A) = \text{rank}([A|B]) = n$ (B) $\text{rank}(A) = \text{rank}([A|B]) < n$
(C) $\text{rank}(A) < \text{rank}([A|B])$ (D) $\text{rank}(A) = 0$

14. The system $AX = B$ with n unknowns has infinitely many solutions if:
(A) $\text{rank}(A) = \text{rank}([A|B]) = r < n$ (B) $\text{rank}(A) = \text{rank}([A|B]) = n$
(C) $\text{rank}(A) \neq \text{rank}([A|B])$ (D) $\text{rank}(A) = n$ and $\text{rank}([A|B]) = n + 1$

15. If $\text{rank}(A) \neq \text{rank}([A|B])$, then $AX = B$ is:
(A) Inconsistent (no solution) (B) Unique solution
(C) Infinitely many solutions (D) Trivial solution

16. If A is a square matrix of order n and $\text{rank}(A) = r < n$, then $\det(A)$ is:
(A) 0 (B) 1 (C) Non-zero (D) r

17. Performing row operations on $[A|I_n]$ until A becomes I_n transforms the right block into:
(A) A^{-1} (B) A^T (C) $\text{adj}(A)$ (D) I_n

18. If P, Q are non-singular such that $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, then $\text{rank}(A)$ is:
(A) r (B) Order of P (C) Order of Q (D) 0

19. Multiplying a matrix by a non-singular matrix:
(A) Preserves matrix rank (B) Reduces rank
(C) Increases rank (D) Makes rank zero

20. The rank of a non-zero column vector of size $n \times 1$ is:
(A) 1 (B) n (C) 0 (D) $n - 1$

21. The rank of a zero matrix $0_{m \times n}$ is:
(A) 0 (B) 1 (C) $\min(m, n)$ (D) Undefined

22. If $A_{3\times4}$ and $B_{4\times3}$ have $\text{rank}(A) = 3$ and $\text{rank}(B) = 3$, maximum rank of AB is:
(A) 3 (B) 4 (C) 12 (D) 1

23. Find the rank of $A = \begin{bmatrix} 2 & 4 & 6 \\ 3 & 6 & 9 \\ 5 & 10 & 15 \end{bmatrix}$.
(A) 1 (B) 2 (C) 3 (D) 0

24. Find the rank of $A = \begin{bmatrix} 3 & 0 & 0 \\ 1 & 4 & 0 \\ 2 & 5 & 6 \end{bmatrix}$.
(A) 3 (B) 2 (C) 1 (D) 0

25. Find the rank of $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 4 & 8 \\ 3 & 6 & 12 \end{bmatrix}$.
(A) 1 (B) 2 (C) 3 (D) 0

26. Find the rank of $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 3 \\ 4 & -1 & 7 \end{bmatrix}$.
(A) 2 (B) 3 (C) 1 (D) 0

27. For what value of k does $A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & k & 3 \\ 3 & 6 & 9 \end{bmatrix}$ have rank 1?
(A) $k = 2$ (B) $k = 4$ (C) $k = 0$ (D) $k = 1$

28. Determine the rank of $A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 6 & 10 & 14 \\ 3 & 9 & 15 & 21 \end{bmatrix}$.
(A) 1 (B) 2 (C) 3 (D) 4

29. Find the rank of $A = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.
(A) 3 (B) 2 (C) 1 (D) 0

30. Find the rank of $A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.
(A) 2 (B) 3 (C) 1 (D) 4

31. Reduce $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ to normal form.
(A) I_2 (B) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (C) 0_2 (D) I_1

32. Find non-singular P, Q such that $PAQ = I_2$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$. Rank of A is:
(A) 2 (B) 1 (C) 0 (D) 3

33. Determine consistency: $x + y + z = 4$, $2x + 3y + 4z = 11$, $x + 2y + 4z = 11$.
(A) Consistent with unique solution (B) Consistent with infinite solutions
(C) Inconsistent (D) Trivial solution only

34. Solve for z in: $x + y + z = 4$, $2x + 3y + 4z = 11$, $x + 2y + 4z = 11$.
(A) $z = 2$ (B) $z = 4$ (C) $z = 3$ (D) $z = 0$

35. Find y in: $x + y + z = 4$, $2x + 3y + 4z = 11$, $x + 2y + 4z = 11$.
(A) $y = 3$ (B) $y = 1$ (C) $y = -5$ (D) $y = 0$

36. Find x in: $x + y + z = 4$, $2x + 3y + 4z = 11$, $x + 2y + 4z = 11$.
(A) $x = -1$ (B) $x = 1$ (C) $x = 0$ (D) $x = 5$

37. For what value of a is $x + y + z = 2$, $x + 2y + 3z = 5$, $x + 3y + az = 9$ INCONSISTENT?
(A) $a = 5$ (B) $a \neq 5$ (C) $a = 0$ (D) $a = 3$

38. System $x+y+z=6$, $x+2y+3z=10$, $2x+3y+4z=16$: find $\text{rank}(A)$ and $\text{rank}([A|B])$.
(A) $\text{rank}(A)=2, \text{rank}([A|B])=2$ (B) $\text{rank}(A)=3, \text{rank}([A|B])=3$
(C) $\text{rank}(A)=2, \text{rank}([A|B])=3$ (D) $\text{rank}(A)=1, \text{rank}([A|B])=1$

39. Determine c such that $x+y+z=5$, $x+2y+3z=9$, $x+2y+cz=12$ is INCONSISTENT.
(A) $c=3$ (B) $c\neq 3$ (C) $c=0$ (D) $c=1$

40. Determine c such that system in above question has UNIQUE solution.
(A) $c\neq 3$ (B) $c=3$ (C) $c=2$ (D) $c=0$

41. Find rank of homogeneous matrix $A=\begin{bmatrix}1&1&1\\2&2&2\\3&3&3\end{bmatrix}$.
(A) 1 (B) 3 (C) 2 (D) 0

42. Number of independent non-trivial solutions for $AX=0$ with $A=\begin{bmatrix}1&1&1\\2&2&2\\3&3&3\end{bmatrix}$ is:
(A) $3-1=2$ (B) 1 (C) 0 (D) 3

43. Find rank of $A=\begin{bmatrix}1&2&3\\2&4&7\\3&6&10\end{bmatrix}$.
(A) 2 (B) 3 (C) 1 (D) 0

44. Find rank of $A=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$.
(A) 2 (B) 3 (C) 1 (D) 0

45. Find k if $A=\begin{bmatrix}1&2\\3&k\end{bmatrix}$ has rank 1.
(A) $k=6$ (B) $k=5$ (C) $k=0$ (D) $k=1$

46. Find rank of $A=\begin{bmatrix}1&2&3\\0&0&0\\0&0&0\end{bmatrix}$.
(A) 1 (B) 3 (C) 2 (D) 0

47. Find rank of $A=\begin{bmatrix}1&0\\0&1\\1&1\end{bmatrix}$.
(A) 2 (B) 3 (C) 1 (D) 0

48. System $2x+3y=5$, $4x+6y=10$ has:
(A) Infinitely many solutions (B) Unique solution
(C) No solution (D) Trivial solution only

49. System $2x+3y=5$, $4x+6y=12$ has:
(A) No solution (B) Unique solution
(C) Infinitely many solutions (D) $x=0, y=0$

50. System $x+2y=3$, $3x+y=4$ has:
(A) Unique solution (B) No solution
(C) Infinite solutions (D) Inconsistent

51. Rank of identity matrix I_4 is:
(A) 4 (B) 1 (C) 0 (D) 2

52. Rank of diagonal matrix $\text{diag}(5, 0, 3, 0)$ is:
(A) 2 (B) 4 (C) 1 (D) 3

53. If $\text{rank}(A_{3\times3}) = 3$, then system $AX = 0$ has:
(A) Trivial solution $X = 0$ only (B) Infinitely many solutions
(C) No solution (D) 2 independent parameters

54. Normal form of $A = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix}$ is:
(A) I_2 (B) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (C) 0_2 (D) I_1

55. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, rank is:
(A) 1 (B) 2 (C) 3 (D) 0

56. If A is 3×3 with $\det(A) = 5$, rank of A is:
(A) 3 (B) 2 (C) 1 (D) 0

57. If A is 3×3 with $\det(A) = 0$ and at least one 2×2 minor non-zero, rank is:
(A) 2 (B) 3 (C) 1 (D) 0

58. System $x + y = 2$, $x - y = 0$ has solution:
(A) $x = 1, y = 1$ (B) $x = 2, y = 0$ (C) $x = 0, y = 0$ (D) No solution

59. System $x + y = 2$, $2x + 2y = 5$ is:
(A) Inconsistent (B) Consistent with unique solution
(C) Consistent with infinite solutions (D) Homogeneous

60. Rank of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is:
(A) 1 (B) 0 (C) 3 (D) 2

Section 3: Linear Algebra – Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem

61. The characteristic equation of a square matrix A is defined by:
(A) $\det(A - \lambda I) = 0$ (B) $\det(A + \lambda I) = 1$
(C) $(A - \lambda I)X = 1$ (D) $\text{trace}(A - \lambda I) = 0$

62. A scalar λ is an eigenvalue of A if and only if there exists a non-zero vector X such that:
(A) $AX = \lambda X$ (B) $AX = \lambda I$
(C) $A + X = \lambda X$ (D) $AX = X$

63. The sum of the eigenvalues of a square matrix A equals:
(A) The trace of A (sum of main diagonal elements) (B) The determinant of A
(C) The rank of A (D) The nullity of A

64. The product of the eigenvalues of a square matrix A equals:
(A) $\det(A)$ (B) $\text{trace}(A)$ (C) $\text{rank}(A)$ (D) 0

65. If λ is an eigenvalue of an invertible matrix A, then an eigenvalue of A^{-1} is:
(A) $\frac{1}{\lambda}$ (B) $-\lambda$ (C) λ^2 (D) λ

66. If λ is an eigenvalue of A, then an eigenvalue of A^k ($k \in \mathbb{N}$) is:
(A) λ^k (B) $k\lambda$ (C) $\frac{\lambda}{k}$ (D) $\lambda + k$

67. The matrices A and A^T have:
(A) Exactly the same eigenvalues (B) Reciprocal eigenvalues
(C) Negative eigenvalues (D) Completely different eigenvalues

68. Eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are:
(A) Orthogonal (B) Linearly dependent (C) Equal (D) Parallel

69. A matrix A has at least one eigenvalue equal to zero if and only if:
(A) A is singular ($\det(A) = 0$) (B) A is non-singular
(C) A is identity matrix (D) $\text{trace}(A) = 0$

70. The Cayley-Hamilton Theorem states that every square matrix:
(A) Satisfies its own characteristic equation (B) Is diagonalizable
(C) Has distinct eigenvalues (D) Is symmetric

71. If the characteristic polynomial of $A_{3\times 3}$ is $\lambda^3 - c_1\lambda^2 + c_2\lambda - c_3 = 0$, by CHT:
(A) $A^3 - c_1A^2 + c_2A - c_3I = 0$ (B) $A^3 + c_1A^2 + c_2A + c_3I = 0$
(C) $A^3 - c_1A^2 + c_2A - c_3 = 0$ (D) $A^2 - c_1A + c_2I = 0$

72. A matrix $A_{n\times n}$ is diagonalizable if and only if:
(A) It has n linearly independent eigenvectors (B) $\det(A) = 0$
(C) All eigenvalues are zero (D) It is upper triangular

73. Eigenvalues of a triangular (upper or lower) or diagonal matrix are simply:
(A) Its main diagonal elements (B) All zeros (C) All ones (D) Sum of row elements

74. If λ is an eigenvalue of A with eigenvector X, then an eigenvalue of $A^2 + 3A - 2I$ is:
(A) $\lambda^2 + 3\lambda - 2$ (B) $2\lambda + 3$
(C) $\lambda^2 + 3$ (D) $\frac{\lambda^2+3\lambda}{2}$

75. Find the eigenvalues of $A = \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix}$.
(A) $3, 7$ (B) $0, 10$ (C) $21, 1$ (D) $3, 3$

76. Find the eigenvalues of $A = \begin{bmatrix} 4 & 1 \\ 0 & 5 \end{bmatrix}$.
(A) $4, 5$ (B) $1, 9$ (C) $20, 0$ (D) $4, 4$

77. Find the trace of $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 2 \\ 3 & 1 & 6 \end{bmatrix}$.
(A) 12 (B) 10 (C) 8 (D) 15

78. The sum of eigenvalues of $A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 2 \\ 3 & 1 & 6 \end{bmatrix}$ is:
(A) 12 (B) 24 (C) 6 (D) 0

79. Find the product of eigenvalues of $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.
(A) 6 (B) 5 (C) 2 (D) 3

80. Find the characteristic equation of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
(A) $\lambda^2 - 5\lambda - 2 = 0$ (B) $\lambda^2 - 5\lambda + 2 = 0$
(C) $\lambda^2 + 5\lambda - 2 = 0$ (D) $\lambda^2 - 4\lambda - 1 = 0$

81. Find the eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$.
(A) $-1, -2$ (B) $1, 2$ (C) $0, -3$ (D) $3, -1$

82. If eigenvalues of A are $2, 3$, find eigenvalues of A^2.
(A) $4, 9$ (B) $2, 3$ (C) $5, 6$ (D) $8, 27$

83. If eigenvalues of A are $1, 4$, find eigenvalues of A^{-1}.
(A) $1, \frac{1}{4}$ (B) $-1, -4$ (C) $1, 16$ (D) $4, 1$

84. If eigenvalues of A are $2, 5$, find eigenvalues of $3A$.
(A) $6, 15$ (B) $2, 5$ (C) $5, 8$ (D) $6, 5$

85. If eigenvalues of A are $1, 2, 3$, find $\det(A)$.
(A) 6 (B) 1 (C) 5 (D) 0

86. If eigenvalues of A are $1, 2, 3$, find $\text{trace}(A)$.
(A) 6 (B) 5 (C) 11 (D) 1

87. If $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, find its eigenvalues.
(A) $5, -1$ (B) $1, 3$ (C) $2, 4$ (D) $4, -2$

88. An eigenvector of $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ corresponding to $\lambda = 5$ is:
(A) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
(C) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ (D) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

89. An eigenvector of $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ corresponding to $\lambda = -1$ is:
(A) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
(C) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (D) $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$

90. If $A^2 - 4A + 3I = 0$, express A^{-1} using CHT.
(A) $A^{-1} = \frac{1}{3}(4I - A)$ (B) $A^{-1} = 4I - A$
(C) $A^{-1} = 3I - 4A$ (D) $A^{-1} = \frac{1}{4}(A - 3I)$

91. If $A^2 - 5A + 6I = 0$, evaluate A^3.
(A) $19A - 30I$ (B) $5A - 6I$ (C) $25A - 36I$ (D) $30A - 19I$

92. Characteristic equation of $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ is:
(A) $(\lambda - 1)^2(\lambda - 3) = 0$ (B) $(\lambda - 2)^3 = 0$
(C) $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$ (D) Both A and C

93. Eigenvalues of matrix in Question mentioned above are:
(A) $1, 1, 3$ (B) $2, 2, 2$ (C) $1, 2, 3$ (D) $0, 1, 4$

94. If eigenvalues of $A_{3\times3}$ are $1, 2, 4$, characteristic equation is:
(A) $\lambda^3 - 7\lambda^2 + 14\lambda - 8 = 0$ (B) $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$
(C) $\lambda^3 - 8\lambda^2 + 14\lambda - 7 = 0$ (D) $\lambda^3 - 7\lambda + 8 = 0$

95. By CHT, for matrix in Question mentioned above, $A^3 =$
(A) $7A^2 - 14A + 8I$ (B) $6A^2 - 11A + 6I$
(C) $8A^2 - 14A + 7I$ (D) $7A - 8I$

96. Inverse of matrix with characteristic equation $\lambda^3 - 7\lambda^2 + 14\lambda - 8 = 0$ is:
(A) $\frac{1}{8}(A^2 - 7A + 14I)$ (B) $\frac{1}{7}(A^2 - 8A + 14I)$
(C) $A^2 - 7A + 14I$ (D) $\frac{1}{14}(A^2 - 7A + 8I)$

97. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, eigenvalues of $A^3 - 2A + I$ are:
(A) $0, 5, 22$ (B) $1, 8, 27$ (C) $1, 4, 9$ (D) $0, 2, 6$

98. If $\det(A - \lambda I) = -\lambda^3 + 6\lambda^2 - 11\lambda + 6$, product of eigenvalues is:
(A) 6 (B) -6 (C) 11 (D) 1

99. If $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$, trace of A is:
(A) 0 (B) 6 (C) -6 (D) 25

100. Determinant of $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$ is:
(A) -25 (B) 25 (C) 0 (D) -7

101. Eigenvalues of $A = \begin{bmatrix} 3 & 4 \\ 4 & -3 \end{bmatrix}$ are:
(A) $5, -5$ (B) $3, -3$ (C) $25, -25$ (D) $0, 5$

102. Eigenvalues of $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ are:
(A) $1, -1$ (B) $1, 1$ (C) $0, 0$ (D) $2, -2$

103. Eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ are:
(A) $i, -i$ (B) $1, -1$ (C) $0, 0$ (D) $1, 1$

104. If A is 2×2 with $\text{trace}(A) = 8$ and $\det(A) = 12$, its eigenvalues are:
(A) $2, 6$ (B) $3, 5$ (C) $4, 4$ (D) $1, 12$

105. If A is 2×2 with $\text{trace}(A) = 5$ and $\det(A) = 6$, its eigenvalues are:
(A) $2, 3$ (B) $1, 6$ (C) $-1, -6$ (D) $0, 5$

106. If eigenvalues of A are $0, 2, 5$, then rank of A is at most:
(A) 2 (B) 3 (C) 1 (D) 0

107. Eigenvalues of I_3 are:
(A) $1, 1, 1$ (B) $3, 0, 0$ (C) $0, 0, 0$ (D) $1, 2, 3$

108. Eigenvalues of zero matrix $0_{3\times3}$ are:
(A) $0, 0, 0$ (B) $1, 1, 1$ (C) Undefined (D) $1, 0, 0$

109. If A is 3×3 with eigenvalues $1, -1, 2$, $\det(A^2)$ is:
(A) 4 (B) -2 (C) 2 (D) 16

110. If A is 3×3 with eigenvalues $1, -1, 2$, $\text{trace}(A^2)$ is:
(A) $1 + 1 + 4 = 6$ (B) 2 (C) 4 (D) 8

111. If A is 3×3 matrix with $\det(A) = 10$, product of its eigenvalues is:
(A) 10 (B) 30 (C) 3 (D) 1

ANSWER KEY

Updated CAT-2 Question Bank · Questions 1–111

Section 1: Differential Calculus – Maxima, Minima & Lagrange Multipliers (Q1–51)

1. A **2.** A **3.** A **4.** C **5.** C

6. A **7.** A **8.** A **9.** A **10.** A

11. A **12.** C **13.** B **14.** A **15.** A

16. A **17.** A **18.** A **19.** A **20.** A

21. A **22.** A **23.** A **24.** A **25.** A

26. A **27.** A **28.** A **29.** A **30.** A

31. A **32.** A **33.** A **34.** A **35.** —

36. A **37.** A **38.** A **39.** A **40.** A

41. A **42.** A **43.** A **44.** A **45.** A

46. A **47.** A **48.** A **49.** A **50.** A

51. A

Section 2: Linear Algebra – Matrix Rank, Inverses, Normal Form & Linear Systems (Q1–60)

1. A **2.** A **3.** D **4.** A **5.** A

6. A **7.** A **8.** A **9.** A **10.** A

11. A **12.** A **13.** A **14.** A **15.** A

16. A **17.** A **18.** A **19.** A **20.** A

21. A **22.** A **23.** A **24.** A **25.** A

26. A **27.** A **28.** A **29.** A **30.** A

31. A **32.** A **33.** A **34.** A **35.** C

36. D **37.** — **38.** A **39.** A **40.** A

41. A **42.** A **43.** A **44.** A **45.** A

46. A **47.** A **48.** A **49.** A **50.** A

51. A **52.** A **53.** A **54.** A **55.** A

56. A **57.** A **58.** A **59.** A **60.** A

Section 3: Linear Algebra – Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem (Q61–111)

61. A **62.** A **63.** A **64.** A **65.** A

66. A **67.** A **68.** A **69.** A **70.** A

71. A **72.** A **73.** A **74.** A **75.** A

76. A **77.** A **78.** A **79.** A **80.** A

81. A **82.** A **83.** A **84.** A **85.** A

86. A **87.** A **88.** A **89.** A **90.** A

91. A **92.** A **93.** A **94.** A **95.** A

96. A **97.** A **98.** A **99.** A **100.** A

101. A **102.** A **103.** A **104.** A **105.** A

106. A **107.** A **108.** A **109.** A **110.** A

111. A

Notes on questions with issues

Section 1, Q35: None of the listed options matches the stationary point. Solving the Lagrange equations gives (x, y, z) = (84/41, 56/41, 126/41), with $\lambda = -168/41$. The question/options should be checked.

Section 1, Q49: The correct classification is **Option A (saddle point)**, but the determinant shown in the option is miscalculated. Here r = 4, s = –4, t = 2, so $rt - s^2 = 8 - 16 = -\mathbf{8}$, not –12.

Section 2, Q37: The system is inconsistent when **a = 7**. This value is not among the listed options, so none of the options is correct.

Section 2, Q34–36: These use the same system; its solution is (x, y, z) = (5, –5, 4), so the answers are Q34 A, Q35 C, Q36 D.

Section 1, Q46: The ratio is x:y:z = 6:2:3, equivalent to 1:1/3:1/2 (Option A as typeset).

A dash (—) means the question has no correct listed option; see the notes above.